Course: BTech **Semester:** 2

Prerequisite: Basic Knowledge of Matrix and Determinants

Rationale : To attain the knowledge of Matrices and apply to the engineering field to check the consistency of the system. The concept of vector space helps to understand programming in field of computer engg.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	Internal Marks			External Marks		
					T	CE	P	T	P	
4	-	-	-	4	20	20	-	60	-	100

SEE - Semester End Examination, **T** - Theory, **P** - Practical

Course Content

W - Weightage (%) , **T** - Teaching hours

Sr.	Topics	W	T
1	**Matrices** Introduction, System of linear equations (Homogeneous and Nonhomogeneous systems) by Gauss Elimination, and LU Decomposition Method, Rank, Eigenvalues and Eigenvectors, Algebraic and Geometric Multiplicity, Diagonalization, Cayley-Hamilton theorem and Applications-coding-decoding, Principal Component Analysis (PCA).	35	21
2	**Vector Space** Vector spaces over the field of real numbers, Elementary Properties of Vector Spaces, Subspaces, Spanning set, Linear independence and dependence, Basis and Dimension.	20	12
3	**Linear Transformation** Definition of Linear Transformation from R^n to R^m Some Standard Linear Transformation, The Null Space & Column Space, The Rank-Nullity Theorem, Matrix of Linear Transformation.	25	15
4	**Inner Product Spaces** Inner-product spaces, norm and its properties, angle between two vectors, orthogonal vectors, orthonormal vector, Gram-Schmidt process, orthonormal basis.	20	12
	Total	100	60

Reference Books

1.	**Elementary Linear Algebra with Applications, (TextBook)** By H. Anton | John Wiley
2.	**Linear Algebra and Its Applications** By G. Strang | Brooks/Cole India
3.	**Advanced Engineering Mathematics** By Erwin Kreyszig | Willey India Education
4.	**Linear Algebra,** By K. Hoffman and R. Kunze, | Prentice Hall
5.	**Calculus** By T.M. Apostol | John Wiley Publication | Vol. II
6.	**Linear Algebra: A Modern Introduction,** By D. Poole, | 2nd Edition, Brooks/Cole

Course Outcome

After Learning the Course the students shall be able to:
1. Understand system of linear algebraic equation and solution techniques using matrix. 2. Exhibit the understanding of vector space and subspace and solve related problems. 3. Compute eigenvalues and eigenvectors and diagonalize the matrix 4. Create orthogonal and orthonormal bases: Gram-Schmidt process and use bases and orthonormal bases to solve application problem.